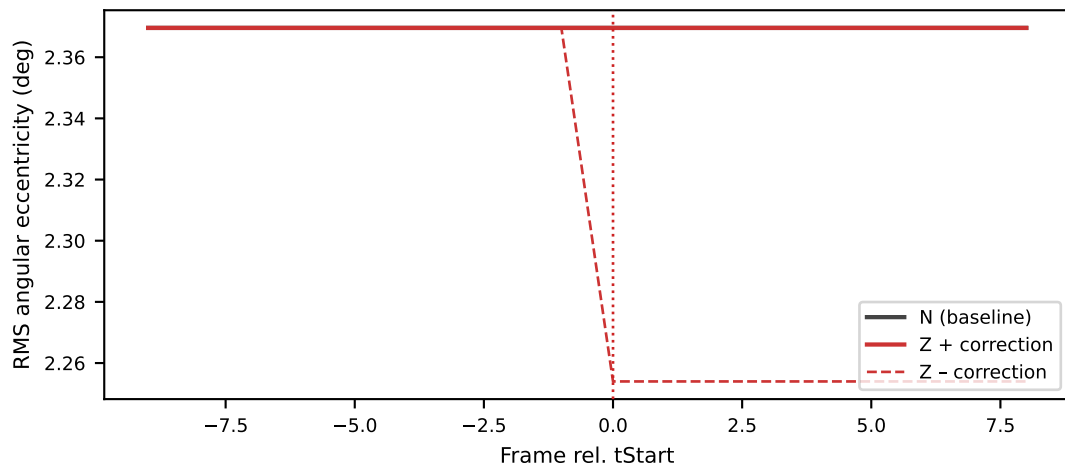
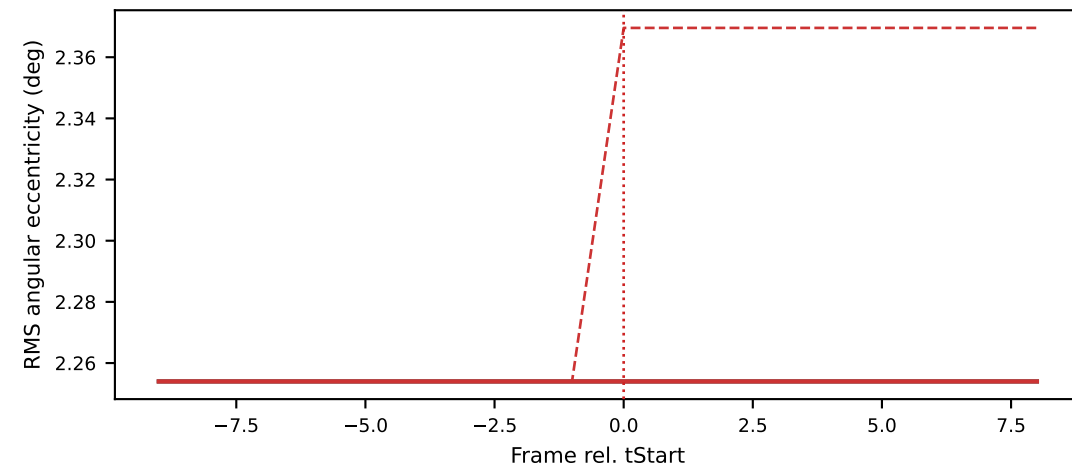


**Angular spread & probe-dot traces — the correct diagnostic for radial expansion**  
**Grey=N Solid-red=Z with correction Dashed-red=Z without correction**  
**Red vertical = tStart. If fix works: solid-red overlaps grey exactly.**

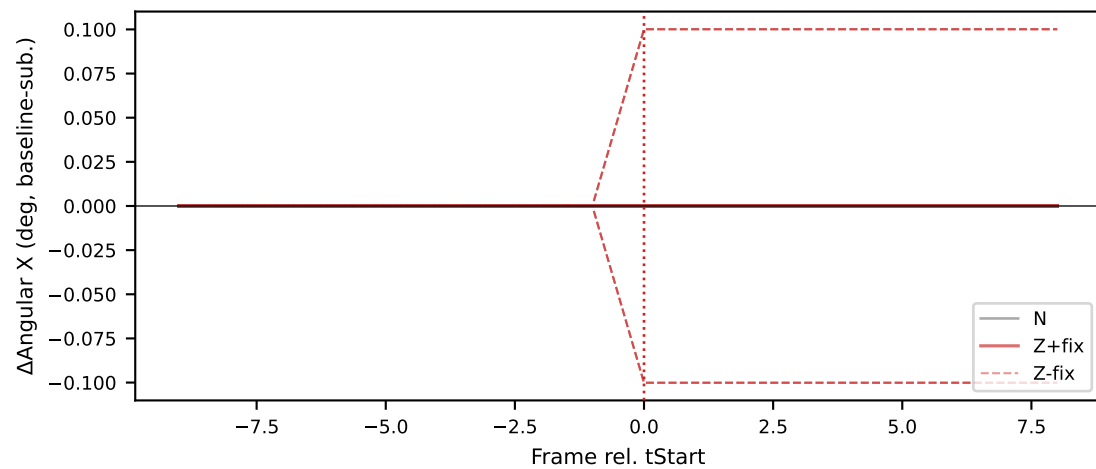
Sub3 RMS spread Near-delayed  
(stationary probe dots — any jump = artifact)



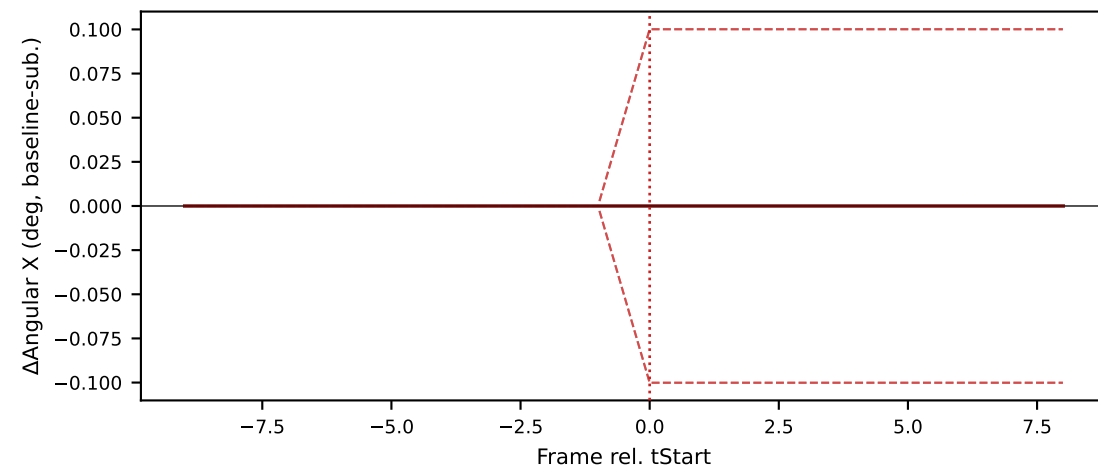
Sub3 RMS spread Far-delayed  
(stationary probe dots — any jump = artifact)



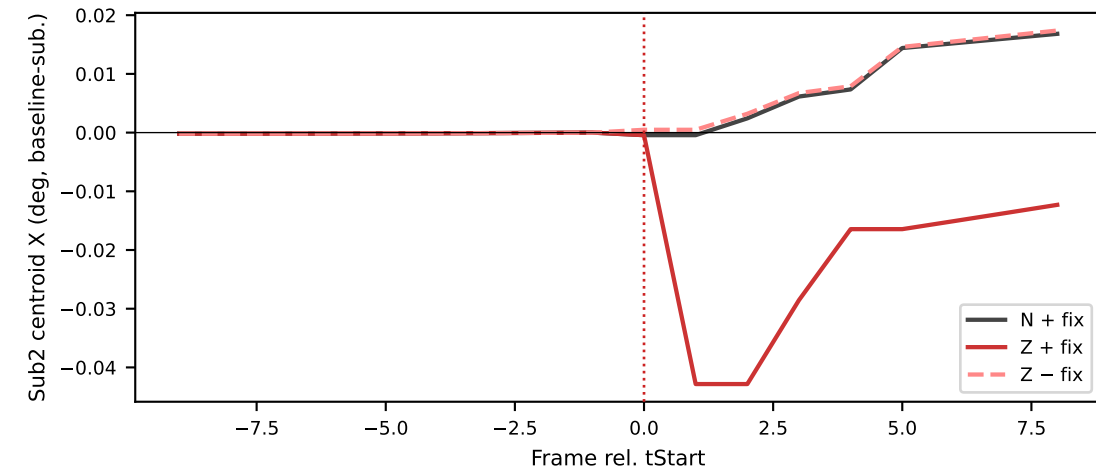
Probe dot angular X Near-delayed  
(9 stationary probes — step = expansion artifact)



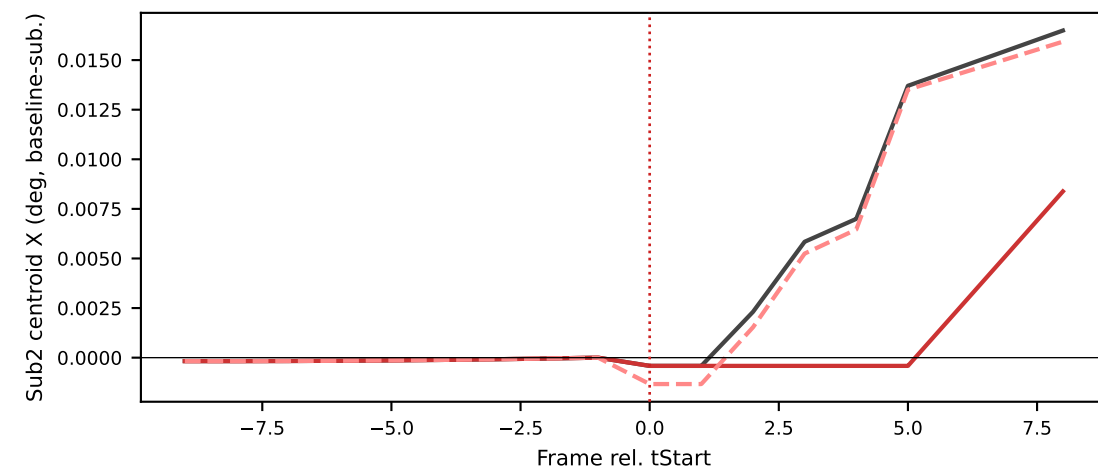
Probe dot angular X Far-delayed  
(9 stationary probes — step = expansion artifact)



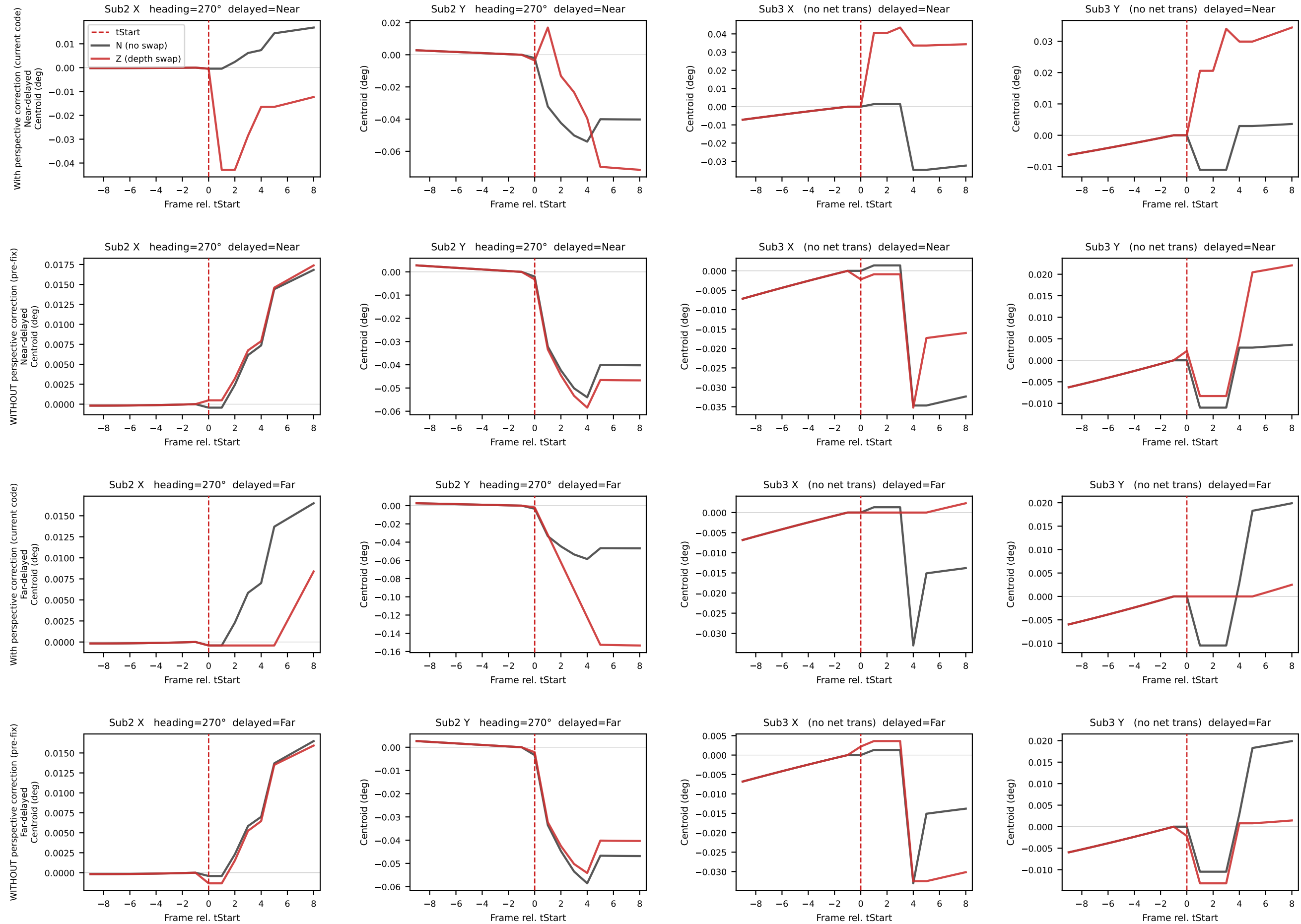
Translation signal Near-delayed heading=270°  
(should be same for N and Z)



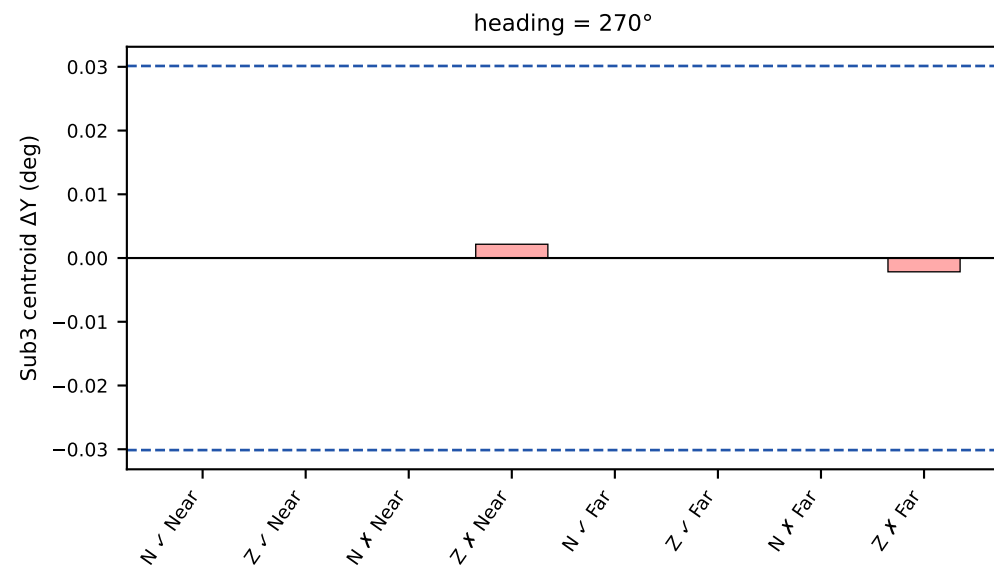
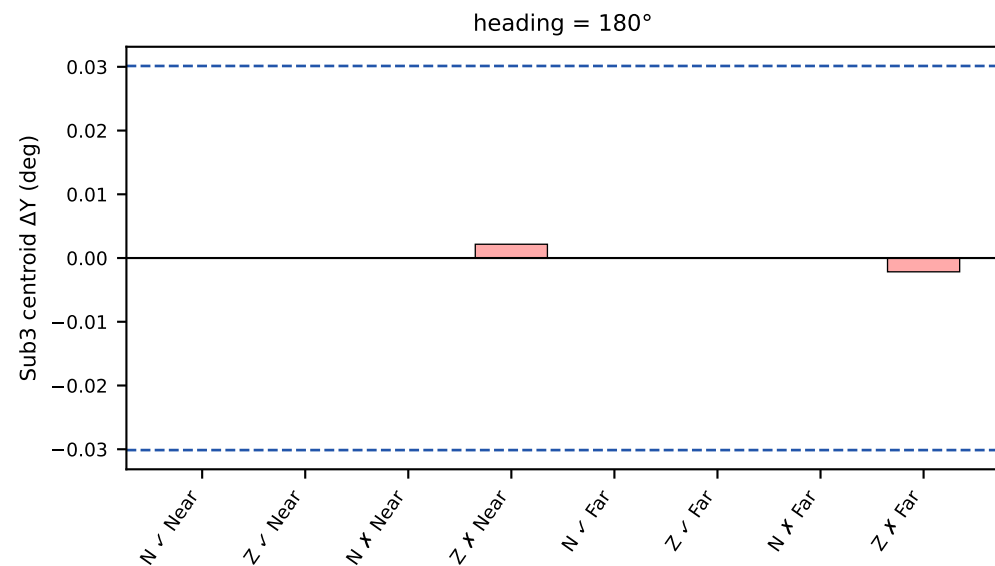
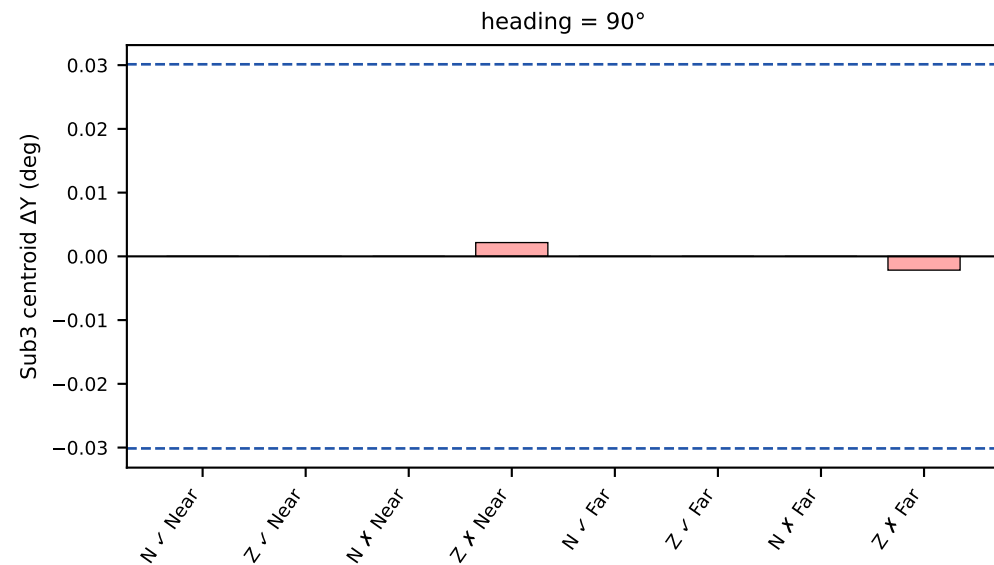
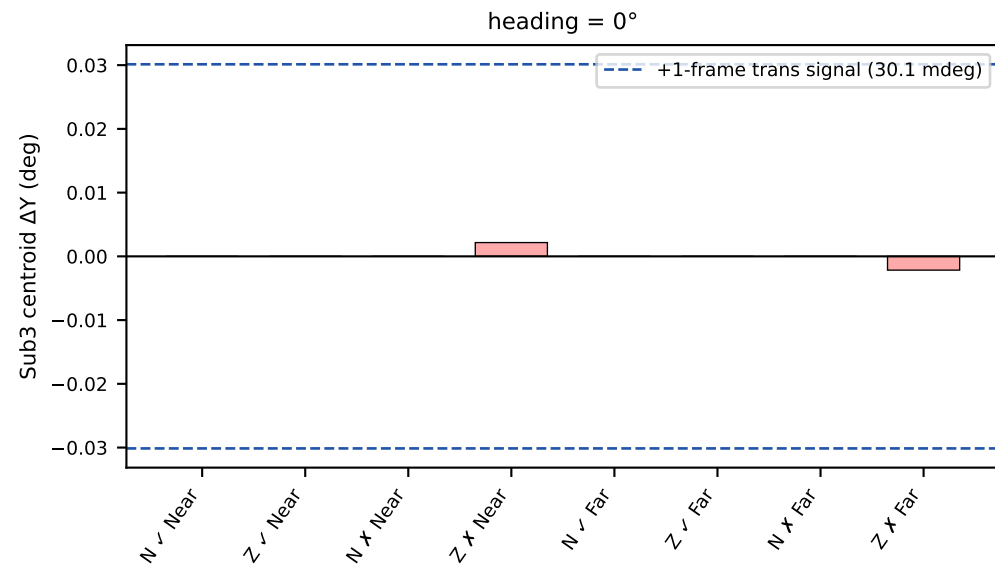
Translation signal Far-delayed heading=270°  
(should be same for N and Z)



**Screen-space centroid traces — 18 frames around tStart**  
**heading=270° CUED | Grey=N(no swap) Red=Z(depth swap) Note: centroid is insensitive to radial expansion (see Page 3 for expansion)**

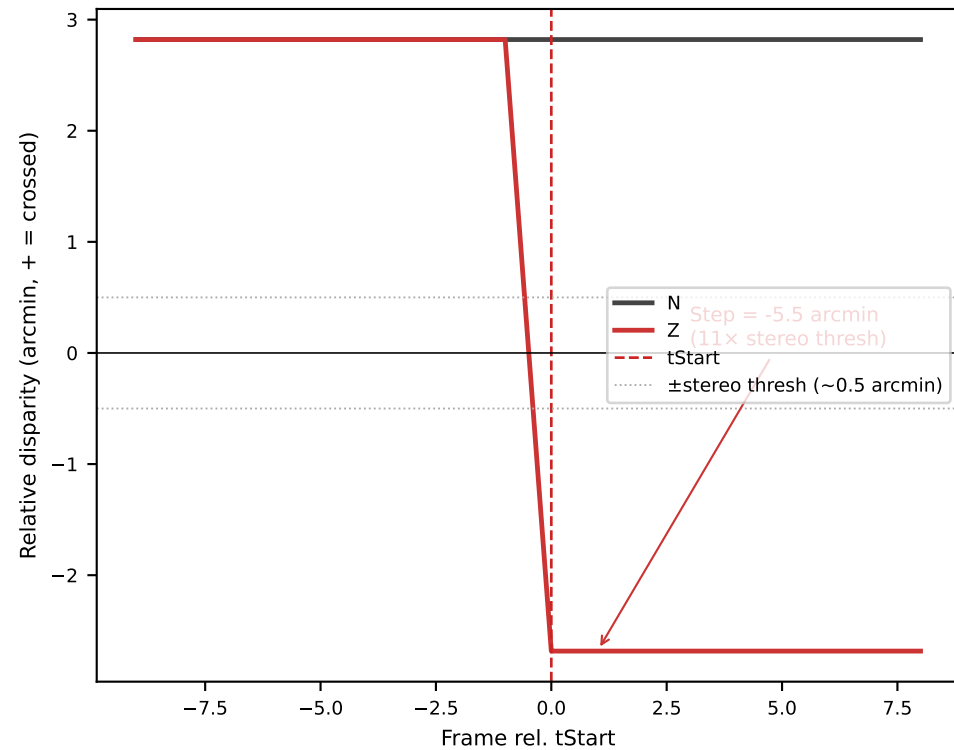


**Sub3 centroid Y-jump at tStart (1 frame) — pure artifact signal**  
**N (no swap) should have  $\approx 0$ ; Z should match N with correction applied**

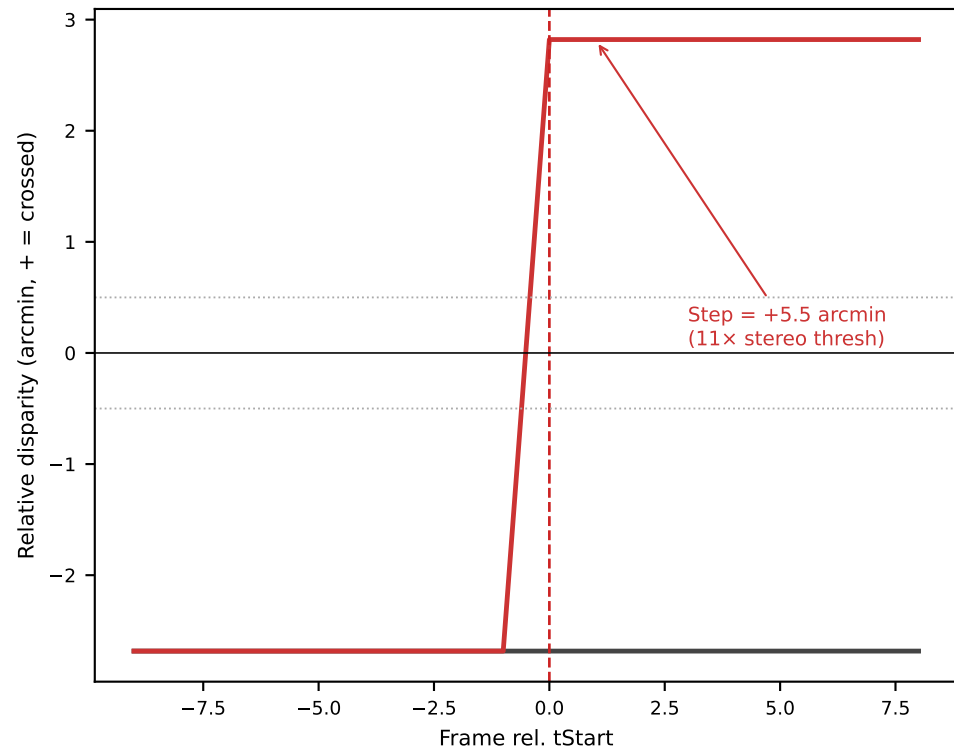


**Relative binocular disparity at depth-swap frame (heading=270°)**  
**This step CANNOT be eliminated — it is the inherent signal of a Z/CZ condition.**  
**Grey=N (no swap, constant disparity) Red=Z (depth swap at tStart = disparity step)**

Near-delayed (with perspective correction)



Far-delayed (with perspective correction)



## headset\_faithful\_traces.py – verification of perspective correction

### Faithfulness to headset rendering

Unity renders with a perspective camera:  $x_{\text{screen}} = x_{\text{world}} / z_{\text{world}}$  (GPU perspective division).  
Previous trace scripts used  $x/\text{viewDist}$  (orthographic) – blind to depth-dependent angular shifts.  
This script computes the same perspective division after running `ApplyDepthOffsets` exactly.

### Key numbers (depthSeparation=0.05 m, viewDist=2.0 m)

Near:  $z=-0.05\text{ m} \rightarrow z_{\text{world}}=1.95\text{ m}$       Far:  $z=+0.05\text{ m} \rightarrow z_{\text{world}}=2.05\text{ m}$

WITHOUT correction – dot at  $2^\circ$  eccentricity, Near→Far:

$$\text{screen\_x before} = \tan(2^\circ) \times 2.0 / 1.95 = 2.052^\circ$$

$$\text{screen\_x after} = \tan(2^\circ) \times 2.0 / 2.05 = 1.952^\circ \quad \Delta = -6.0 \text{ arcmin (inward contraction)}$$

At aperture edge ( $3.5^\circ$ ):  $\Delta = -10.5 \text{ arcmin} \rightarrow$  apparent velocity =  $13.1^\circ/\text{sec}$  ( $5.8\times$  signal)

WITH correction –  $\text{angularScale} = 2.05 / 1.95 = 1.0513$ :

$$\text{screen\_x after} = \tan(2^\circ) \times 2.0 \times 1.0513 / 2.05 = 2.052^\circ \quad \Delta = 0.00 \text{ arcmin} \quad \checkmark$$

INHERENT disparity step (cannot be fixed – this IS the depth-swap):

$$\text{Near: relative disparity} = \text{IPD} \times (1/1.95 - 1/2.0) = +2.76 \text{ arcmin (crossed)}$$

$$\text{Far: relative disparity} = \text{IPD} \times (1/2.05 - 1/2.0) = -2.55 \text{ arcmin (uncrossed)}$$

$$\text{Step at tStart (Near→Far): } \sim 5.3 \text{ arcmin} \approx 11\times \text{ stereo threshold (0.5 arcmin)}$$

This is what produces the residual "pop" percept after all code fixes.

### How to read Page 1 (expansion diagnostic)

Row 0: RMS angular eccentricity of 9 stationary probe dots.

N (grey) = flat. Z without correction (dashed) = drops  $\sim 5\%$  at tStart.

Z with correction (solid red) = flat, overlapping grey. Artifact eliminated.

Row 1: Individual probe dot angular X positions (probes on  $3\times 3$  grid at  $\pm 2^\circ$ ).

Same result. Dashed shows the inward/outward step; solid shows clean.

Row 2: Sub2 translation signal (centroid X) – should ramp equally for N and Z.

How to use: for any future depth-swap experiment, run this script first.

Page 1 Row 0: solid-red must overlap grey. Page 4: disparity step magnitude.